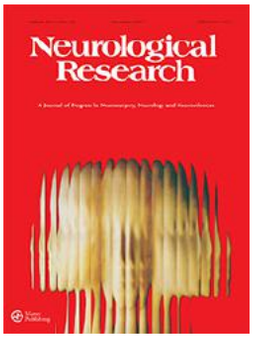

Neurological Research

A Journal of Progress in Neurosurgery, Neurology and Neurosciences

ISSN: 0161-6412 (Print) 1743-1328 (Online) Journal homepage: http://www.tandfonline.com/loi/yner20

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To cite this article: Ramazan Üstün & Peray Ayhan (2018): Regenerative activity of *Hericium erinaceus* on axonal injury model using *in vitro* laser microdissection technique, Neurological Research, DOI: 10.1080/01616412.2018.1556494

To link to this article: https://doi.org/10.1080/01616412.2018.1556494

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https://doi.org/10.1080/01616412.2018.1556494

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Regenerative activity of *Hericium erinaceus* on axonal injury model using *in vitro* laser microdissection technique

Ramazan Üstün[a,b] and Peray Ayhan[b]

[a]Department of Physiology, School of Medicine, Van Yüzüncü Yıl University, Van, Turkey; [b]Neuroscience Research Unit, School of Medicine, Van Yüzüncü Yıl University, Van, Turkey

ABSTRACT

Objective: Peripheral nerve injury (PNI) is an important global health problem. Nerve Growth Factor (NGF) plays crucial role in the survival, growth, and maintenance of various neurons in the mammalian nervous system, human included. *Hericium erinaceus* (HE), an edible and medicinal mushroom, has been extensively studied for its neuroprotective properties. In this study, the neuroprotective and neurotogenic effects of HE and NGF were compared on mouse PNI model by using a laser microdissection technique.

Methods: Neuronal cultures were prepared from dorsal root ganglia (DRG) of 6–8 week aged mice, pretreated them with phosphate-buffered saline (PBS), NGF, HE, or the combination of NGF and HE. To model axonal injury *in vitro*, axons were cut (axotomy) with a microscope-controlled laser beam. Axotomized neurons were imaged under the microscope. Axotomized neurons' survival ratios were calculated using the propidium iodide (PI), which is a red-fluorescent nuclear dye. Their axon lengths were measured using the AxioVision 4.8 software.

Results: Although both HE and NGF have neuroprotective and regenerative effects on axotomized peripheral sensory neurons, HE exhibits a higher neuroprotective activity compared to the NGF. The combination of HE and NGF maximizes axonal regeneration ability of axotomized neurons.

Conclusion: HE has capabilities of preventing the death of neurons and regenerating their axons in the experimental axonal injury model. Our findings provide experimental evidence that HE may serve as a neuroprotective and regenerative candidate for treating peripheral nerve injuries. Present study warrants further investigation of HE as a potential natural compound to remedy PNI.

ARTICLE HISTORY
Received 9 October 2018
Accepted 3 December 2018

KEYWORDS
Hericium erinaceus; NGF; DRG neurons; axotomy; regeneration

1. Introduction

Traumatic peripheral nerve injuries (PNI) are a major global health problem for today's modern society and occur in 2.8% of patients who are affected by multiple trauma [1]. Severe traumas to peripheral nerves can also lead to permanent neurological deficits, including reinnervation failure and the development of chronic pain [2]. Because, due to the fact that chronically denervated Schwann cells within the distal nerve lose capability to proliferate, they cannot provide structural and trophic support needed for axon regeneration [3]. The regeneration of injured nerves depends on the presence of living Schwann cells at the injury site [4]. Therefore, recovery of injured axons relies on neuronal and non-neuronal cells, including immune cells [5]. For regeneration to happen, axons must have intrinsic capability to regrow, the distal environment must support regrowth, and the target tissues must accept new axons [6]. Although a series of regenerative strategies have been developed for the treatment of PNI, including microsurgical repair, pharmacotherapy, and physical rehabilitation, pharmacotherapy has been the most widely used approach for neuro-rehabilitation [1].

Nerve Growth Factor (NGF) is a neurotrophic factor known for its significant roles in the development, survival, outgrowth, and maintenance of different neuron types in the mammalian nervous system [7]. In the peripheral nerve system (PNS), Schwann cells express NGF that support survival of injured neurons and axonal regrowth [4,8,9]. It was reported exogenous mouse NGF treatment might relieve diabetic peripheral neuropathy through increasing the NGF expression in dorsal horn and dorsal root ganglia (DRG) in mouse diabetic peripheral neuropathy model [10]. It has also been shown that administration of topical NGF induces healing action on human cutaneous, corneal and pressure ulcers, glaucoma, maculopathy, and retinitis pigmentosa suggesting a therapeutic potential of NGF in human ophthalmology and cutaneous ulcers [11]. The efficacy of NGF is hindered by the poor pharmacokinetic profile and its inability to cross the blood–brain barrier [12]. Hence, the potential medicinal values of culinary and medicinal mushrooms have attracted the researchers for their pharmacological activities that mimic the NGF activity in the prevention of neurodegenerative diseases [13].

CONTACT Ramazan Üstün ramazanustun@yyu.edu.tr Department of Physiology, School of Medicine, Van Yüzüncü Yıl University, Zeve Yerleşkesi, Tuşba, Van 65080, Turkey

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Mushrooms are generally a good source of nutrients and health-promoting compounds [14]. More than 2000 species of edible and/or medicinal mushrooms have been identified [14]. In recent decades, the medicinal properties of these mushrooms have become an attractive area of research [15–18]. Among these, the culinary and medicinal *Hericium erinaceus* (HE), also known as Lion's Mane Mushroom or Yamabushitake appear to stand out as an exceptional health-promoting species [14]. HE is a well-known mushroom that has been commercially grown in China, consumed as a good source of nutrients in Japan and China, and used as beneficial medicines without harmful effects in traditional Chinese medicine [19–22]. This mushroom contains physiologically significant components, such as β-glucan polysaccharides, diterpenoid compounds, steroids, and other functional ingredients. HE has been extensively documented and possess a range of therapeutic properties, such as hypolipidemic [15,19], anti-cancer [20], anti-microbial [23], anti-diabetic [24], gastroprotective [25], hepatoprotective [26], anti-hyperlipidemic [27], anti-oxidant [28,29], neuroprotective [30–32], immunomodulating [15,16]. HE's fruiting bodies and mycelia contain health beneficial bioactive components. These components are divided into two groups according to molecular weight: Erinacines and hericenones, which are representative of small molecules with significant neurotrophic and neuroprotective effects and the ability to cross blood–brain barrier [14,30]. Previous reports have revealed that upregulation of TrkA and/or Erk1/2 phosphorylation by HE agreed well with their potency to promote neurite outgrowth [30,33]. Therefore, HE may potentiate NGF-induced neurite outgrowth and neuronal survival.

The purpose of this experiment is to compare the regenerative effect of HE and NGF on mouse PNI model by employing the laser microdissection technique and Zeiss live cell imaging microscopy. A dissociated DRG neuronal cultures are widely used as a model system to investigate neuronal intrinsic regenerative capacity and nerve regeneration *in vitro*, mimicking *in vivo* process at the injury site [34]. Therefore, we evaluated the protective and regenerative effects of HE and NGF using the mouse PNI model *in vitro*.

2. Materials and methods

2.1. Drugs and chemicals

Ketamine hydrochloride was purchased from Pfizer (İstanbul, Turkey). NeuroBasal A medium (NBA), B27 supplement, GlutaMAX, calcein-acetoxymethylester (calcein AM) were purchased from Life Technologies (Grand Island, NY, U.S.A.). Antibiotic-antimycotic solution, RPMI-1640 medium, fetal bovine serum, Hanks' balanced salt solution, poly-L-lysine hydrobromide, Laminin, Trypsin solution, Trypsin inhibitör, collagenase, Deoxyribonuclease I (DNAz), propidium iodide (PI), NGF, phosphate-buffered saline (PBS) tablets were all purchased from Sigma-Aldrich (St. Louis, MO, U.S.A.) and HE was purchased from Agroma (Denizli, Turkey).

2.2. Experimental animals

All of the procedures were approved by the local animal experiment ethics commission, in accordance with the European Community Council Directive 86/609/ECC for the care and use of laboratory animals. DRG primary cultures were prepared from six to eight-week-old Male Balb-C mice purchased from the Van Yüzüncü Yıl University Experimental Animal Production and Research Center. Every effort was made to minimize the suffering of animals and to reduce the number of animals used. Mice were sacrificed by cervical transection under deep anesthesia after an intraperitoneal injection of ketamine (100 mg/kg, Ketalar, Pfizer) [35].

2.3. Preparation of primary culture of DRG neurons

The primary cultures of DRG neurons were prepared from mice as previously reported in detail [35]. DRG neurons were gently dissociated by enzymatic digestion and mechanical trituration. The culture medium was replaced to Neuro Basal-A medium supplemented with 2% B27 containing 2 mM GlutaMAX-I, 100 mg streptomycin, 100 units of penicillin and 250 ng/ml amphotericin-B, and 100 U/ml collagenase, which is recommended for primary cultures of DRG neurons [36]. Resulting single cells were seeded on to 35 mm glass-bottomed petri dishes which had been previously coated with poly-L lysine (18 μg/mm^2, 3 h at room temperature) and laminin (40 ng/mm^2, overnight) and maintained at 37°C in a humidified atmosphere containing 5% CO_2 and 95% air [37]. The preparations were kept in the incubator for 2 h to let the neurons attach to the bottom of the dish, after which they were gently washed with warm NBA-B27 to remove unattached cells and replaced with NBA-B27 and placed back in the incubator. The DRG neurons were maintained at 37°C in 100% humidity and supplied with 5% CO_2 for 48 h [35,36].

2.4. Preparation of hot aqueous extract of Hericium erinaceus

HE's fresh fruiting bodies were sliced into 3-cm square pieces and gently boiled with triple-distilled water at a ratio of 1:10 (w/v) under agitation at 60 ± 2°C for 30 min. The boiled mushroom was left covered for 30 min. Residues were then removed by filtration through absorbent cotton gauze cloth and

centrifuged at 10,000 rpm for 30 min at 4°C. Supernatants were filtered through Whatman Number 1 filter paper to remove insoluble matter. The sample was freeze-dried (Christ Alpha 1–4 LD Freeze dryer) and the hot aqueous extract (HE) powder was stored at 4 ± 2°C until further use [32].

2.5. Experimental design and treatment of DRG neuron cultures

The two-day-old cultures of DRG neurons were randomly allotted into five experimental groups: control group, axotomy group, NGF (50 ng/ml) treatment group, HE (25 µg/ml) treatment group, NGF (50 ng/ml) + HE (25 µg/ml) treatment group. Before axon transection, the control and axotomy groups were pretreated with PBS (25 µl/ml) for 48 h [38]. The treatment groups were pretreated with NGF or HE or NGF + HE for 48 h. The experiment was repeated four times for each one test group. The axon cut (Axotomy) was performed in a laser microdissection system with Ultra Violet (UV) Laser Unit Equipment [35,37,39]. The longest and thickest neurite among many neurites of each neuron is assumed to be the axon [40] (Figure 3, yellow arrowheads).

2.6. Laser microdissection

The laser microdissection procedure was performed as previously reported [35]. After 48 h of incubation, healthy neurons elongating axon at least 150 µm for axotomy were selected on a computer controlled inverted microscope with a stage incubator that provides a physiological atmosphere (Cell Observer; Zeiss, Oberkochen, Germany) (Figure 3, white arrows). Their coordinates were registered and relating marks were put with a pen on the lid of the dish to serve as references for the following procedures. Neurons showing indications of stress, such as deformed membranes, blebs, vacuoles or beaded neurites during selection were excluded from the study [37]. In order to ensure cell viability, PI (7.5 µM) was added to the medium, and absence of nuclear staining in the selected neurons was confirmed [36]. The axons to be axotomized were marked on the microscopic images, and exact points of axotomy at 150 µm distance from the perikaryon were determined [37,39]. For the control group, an empty area which is 25 µm away from the selected neuron's body was determined. Axon length measurements were done with the standard curve measurement tool of AxioVision 4.8 software before axotomy [41]. Axotomies were performed with a laser microdissection system (PALM Microbeam, Zeiss, Germany), which consists of an inverted microscope (Zeiss Axiovert 200) and an attached UV laser unit, which operates at 337 nm and produces 1–30 pulses per second, each lasting 3 ns and releasing approximately 300 mJ energy. The axotomies were made with a 63× dry phase contrast objective (LD Achroplan N.A. 0.75, Zeiss Germany). All axotomies were performed using exclusive software that can manage the laser beam, the microscope, and the attached CCD camera. The laser was shot at an intensity of 65% (relative value) and focal length of 21 (relative value) for a period of approximately one sec for each axotomy [35,37]. A total of 660 neurites were axotomized. There were no deterioration other parts of the axotomized-neurons [35]. The microscopic images were taken immediately after axotomy.

2.7. Axotomized neurons's survival ratio and axon length assay

After 48 h pretreatment to evaluate the neuronal cell death and survival, PI (7.5 µM) and calcein-acetoxymethylester (calcein AM) (1 µM) dyes were added to the culture medium [36]. The red fluorescent staining determines neuronal and non-neuronal cell death by PI uptake assay. The cells with bright red fluorescence under the fluorescence microscope are considered dead cells. The neurons unstained with PI are assumed alive. Calcein AM shows the living neuronal cell bodies/neurites as bright green under the fluorescence microscope. The cultures were then transferred to a computer-controlled Cell Observer Microscopy System. The phase-contrast and fluorescence images of the cells were obtained under the Cell Observer Microscopy System. AxioVision 4.8 software was used to analyze the images obtained from these experiments. Axon lengths of axotomized alive neurons measured at 48 h pretreatment. Finally, survival ratios of neurons and the average axon length after 48 h was calculated [37].

2.8. Statistical analysis

Descriptive statistics were presented as means and standard error of neurite length and count and percentages of live cells. Kruskal-Wallis test was performed to compare groups for the axon length. Also, Z test was used for comparison of the live cell ratio for the groups. The statistical significance level was considered as 5%, and statistical analysis of all data was performed by using MINITAB v.14 statistical software.

3. Results

3.1. Viability of DRG neurons under in vitro culture conditions

To calculate axotomy-induced neuron deaths accurately, we wanted to confirm that all the selected DRG neurons by PI uptake assay before axotomy were alive and healthy [36]. The cells with bright red fluorescence are dead cells (Figure 1, purple arrow). The absence of nuclear PI staining in the selected

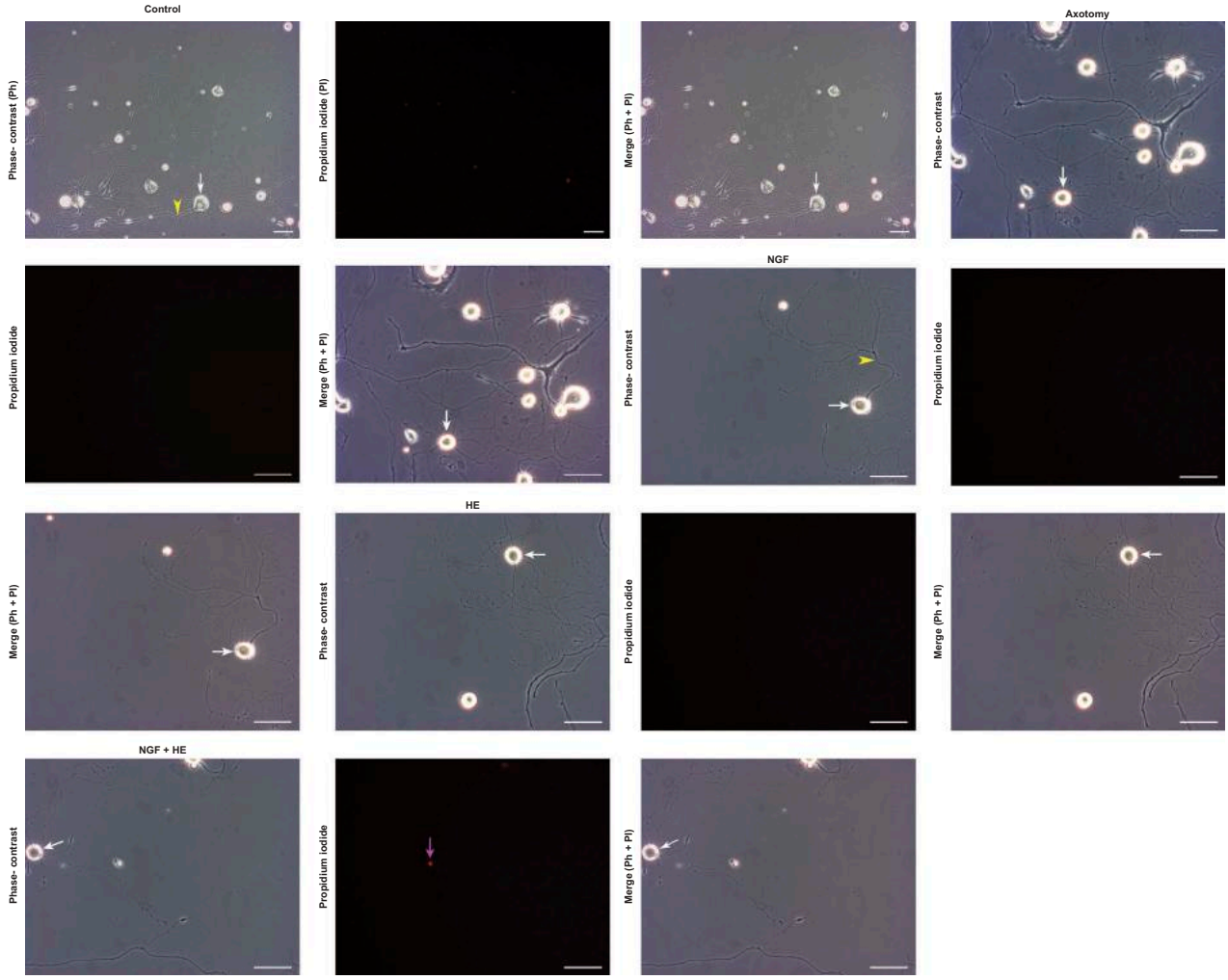

Figure 1. Pretreatment. Phase contrast (Ph), propidium iodide (PI) and merge (Ph + PI) microscopic images before treatment in all groups. DRG neurons were alive and healthy before axotomy (white arrow). These neurons elongated their axons (yellow arrowhead). Ph imaging makes living, unstained cells visible. PI shows the dead cell's nucleus as bright red under the fluorescence microscope (purple arrow). The merged image shows the combination of Ph and PI images. White arrows indicate the neurons selected to be axotomized with the laser beam. Scale bar = 50 µm.

neurons for axotomy was confirmed with fluorescent image. The images indicated that the neurons were alive and healthy before axotomization (Figure 1, white arrows).

3.2. The effects of axotomy on DRG neurons

DRG neurons began the neurite outgrowth within the first 2 h of incubation. The axons reached an average length of 263 µm at around 48 h of *in vitro* culture (Figure 2). Before axotomy, neurons were healthy (Figure 3, white arrows), the integrity and length of the axons were convenient for axotomy (Figure 3, yellow arrowhead). The axons in the control group were not axotomized. The laser beam was shot in an empty area at 25 µm away from the neuron's body. After axotomy, in both the axotomy and experimental groups, the axons divided into two parts; distal parts and proximal parts (Figure 3, red arrowhead). In the axotomy group, the proximal and distal segments lost their integrity and resulted in beading, fragmentation, disappearance (Figure 4, purple arrowhead). Neuronal cell deaths have rapidly followed the fragmentation (Figure 4, purple arrow). Death of a neuron axotomized forming synapse and/or contacting with other cells led to the death of other neuronal and non-neuronal cells (e.g. Schwann and satellite cells) (Figure 4, red arrow).

3.3. Effect of NGF and Hericium erinaceus on viability of axotomized DRG neurons

After 48 h post-axotomy, neuronal cell deaths were determined by the red fluorescent dye stained nucleus by the uptake of PI (Figure 4, purple arrow). The green fluorescent staining determined the viability of neuronal cells by calcein-AM uptake (Figure 4, white arrow). Survival ratios of axotomized neurons were determined as 31% (31/100) for axotomy group, 60% (61/102) for NGF treatment group, 91% (91/100) for HE treatment group, 94% (98/105) for NGF+HE treatment group, and 96% (100/104) for control group (Figure 5). These data suggested that although

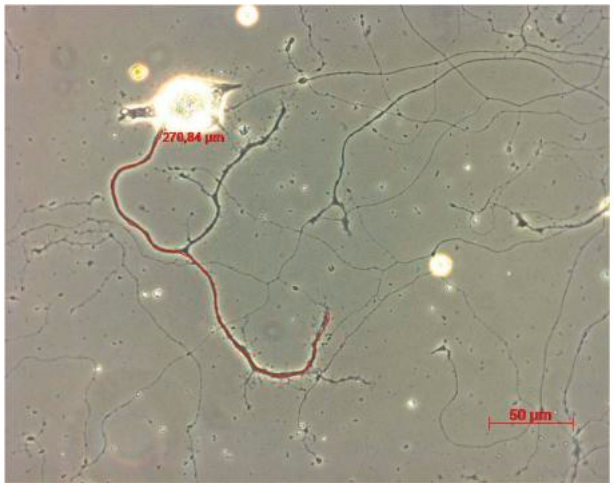

Figure 2. Measurement of axon length with the standard curve measurement tool of AxioVision 4.8 software before axotomy after 48 h of incubation. To avoid damaging the cell body, the longest axon away from the cell body were selected for axotomy.

both NGF and HE significantly increased the survival rate of injured neurons compared to the axotomy group ($p < 0.05$), HE or its combination with NGF had superior activity compared to the NGF in response damage ($p < 0.05$). There were no significant difference among HE, NGF + HE, and control groups. Collectively our studies provide evidence that HE may have a superior neuroprotective activity.

3.4. Effect of NGF and Hericium erinaceus on the axon length of axotomized DRG neurons

The axotomized axons' regrowth is considered to be one of the hallmarks of regeneration [42]. Firstly, the axotomized-neurons quickly retracted axons back to the cell body due to injury stress, overcame the axotomy stress in 24 h, and continued to elongate injured axon rapidly (Figure 4, yellow arrowhead). Axon lengths of axotomized alive neurons were measured at 48-h pretreatment. The average axon lengths in all the group were calculated. The average axon lengths were 6.9 µm for axotomy group, 32.35 µm for NGF treatment group, 32.22 µm for HE treatment and 63.08 µm for the NGF+HE combination group (Figure 5). In the control group, the average length of non-axotomized axons was measured to be 363 µm at 48-h after treatment. Based on our data, both the NGF and HE treatment groups showed significantly longer axon lengths compared to the axotomy group in response to axotomization ($p < 0.05$). When single treatment groups (NGF treatment group and HE treatment group) are compared between themselves, the average axon lengths have the same renewal capacity as it is the same. Interestingly, however, the combination of NGF and HE resulted in significantly longer axon lengths compared to individual compounds or the axotomy group ($p < 0.05$). Consistent with their superior neuroprotective activity, the

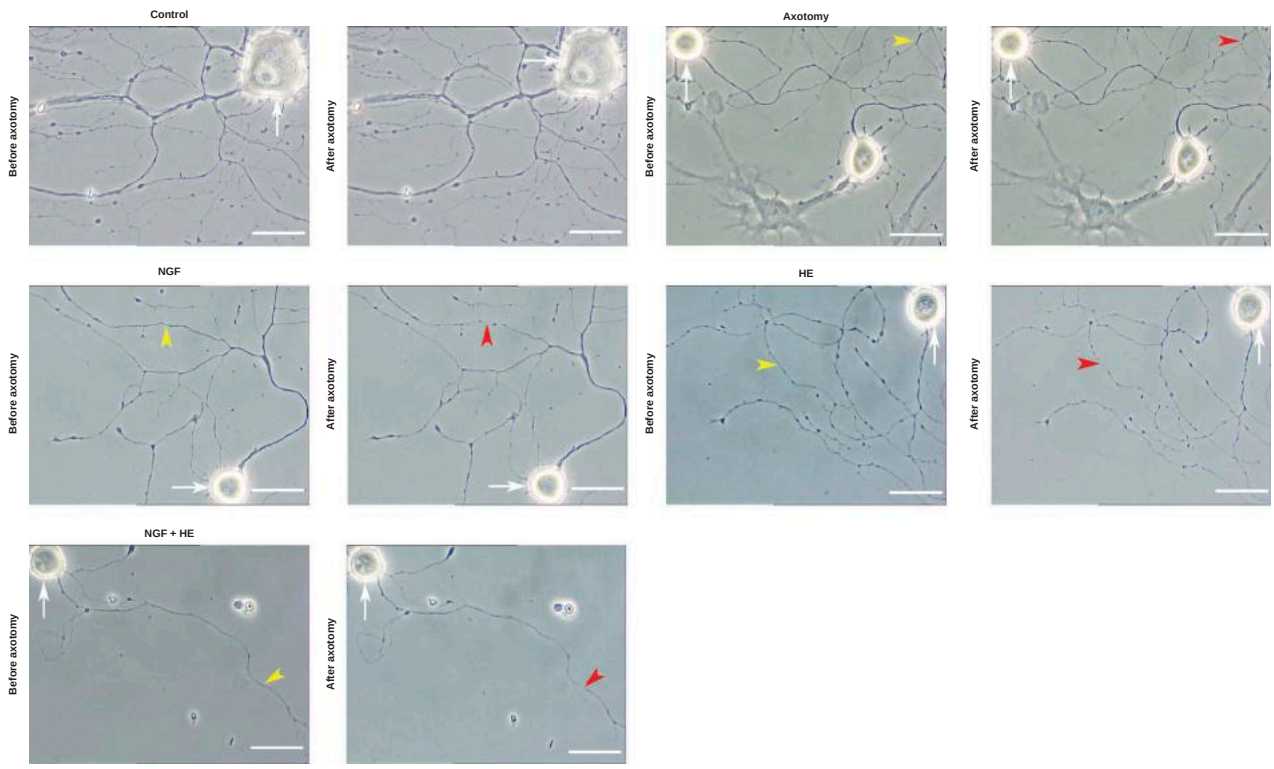

Figure 3. **Axotomy with laser beam**. Axons of alive neurons injured by the laser beam. Microscopic images have been taken immediately before and after axotomy. White arrows indicate alive neurons selected for axotomy. Yellow arrowheads show the integrity of axons that would be axotomized by the laser beam, and red arrowheads show that the axons damaged and divided into two parts. Scale bar = 25 µm.

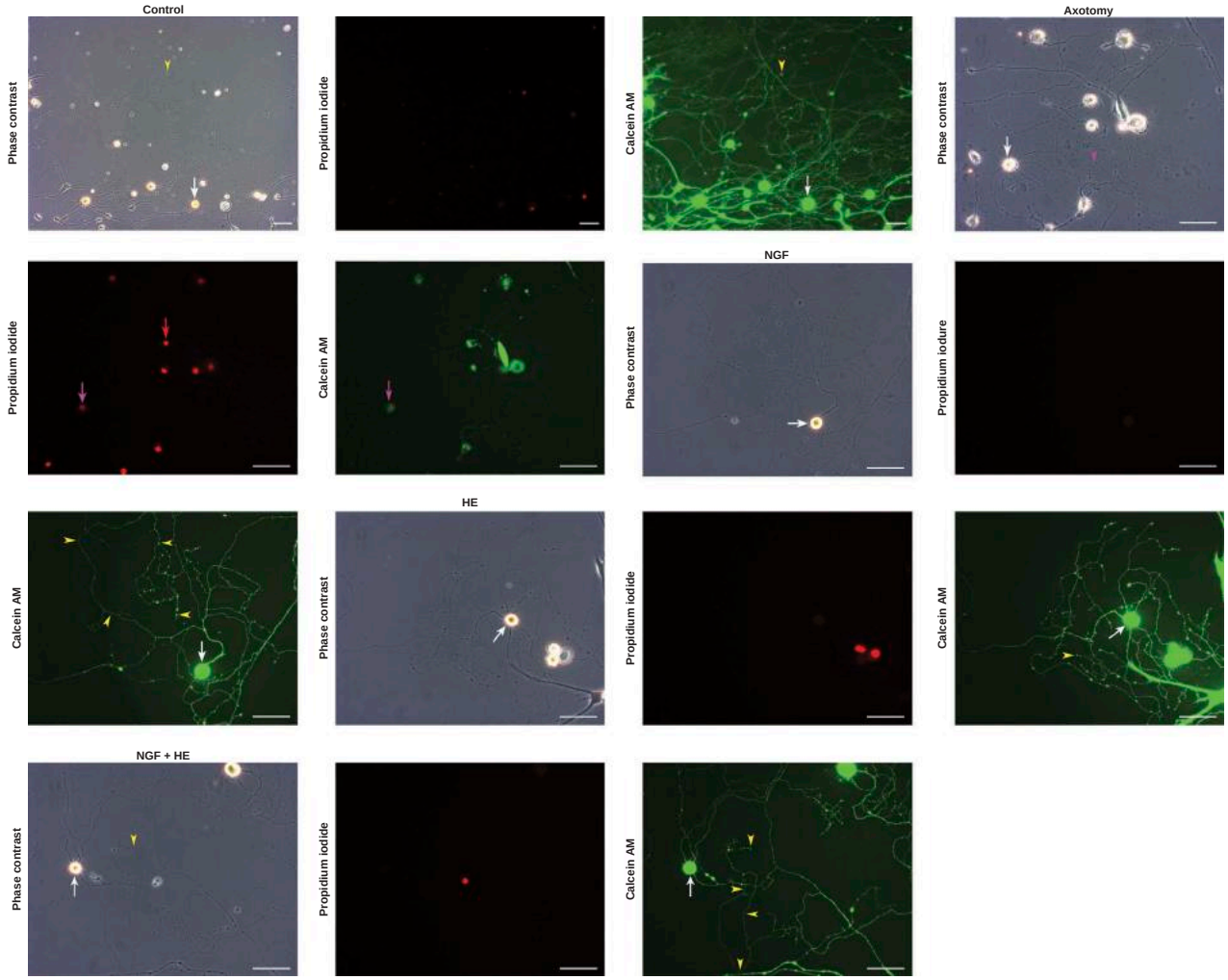

Figure 4. 48 h post-axotomy. Ph, PI and Calcein AM microscopic images of DRG neurons after 48 h from axotomy. Ph imaging makes living, unstained cells visible. PI shows the dead cell's nucleus as bright red under the fluorescence microscope. Calcein AM shows the living neuronal cell bodies/neurites as bright green under the fluorescence microscope. Neurons experienced axotomy stress. White arrows indicate alive neurons with axotomy injury. Yellow arrowheads show the axons' outgrowth. Purple arrows indicate dead neuron with axotomy injury. Purple arrowhead indicates the axons' fragmentation. Red arrow indicates other dead cells contacting with axotomized-neuron. Scale bar = 50 µm.

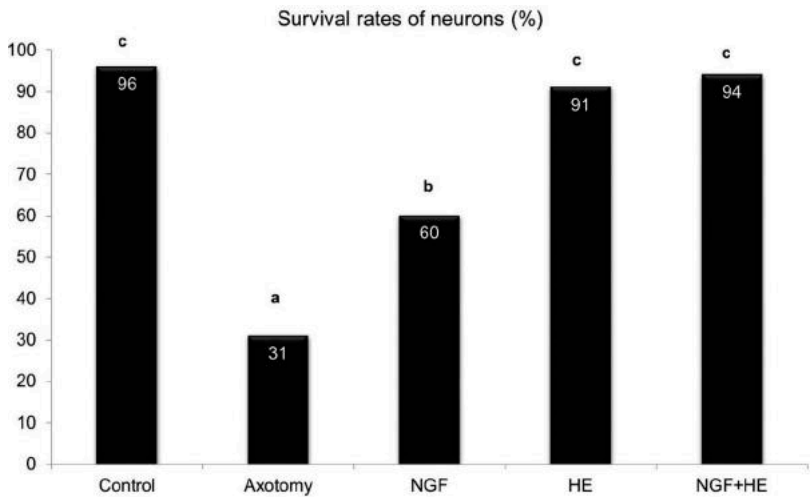

Figure 5. NGF and HE prevent deaths of DRG neurons axotomized. Survival rates of neurons for all the groups 48 h after axotomy injury. The experiment was repeated four times for each one test group. Different lower cases represent statistically significant differences ($p < 0.05$).

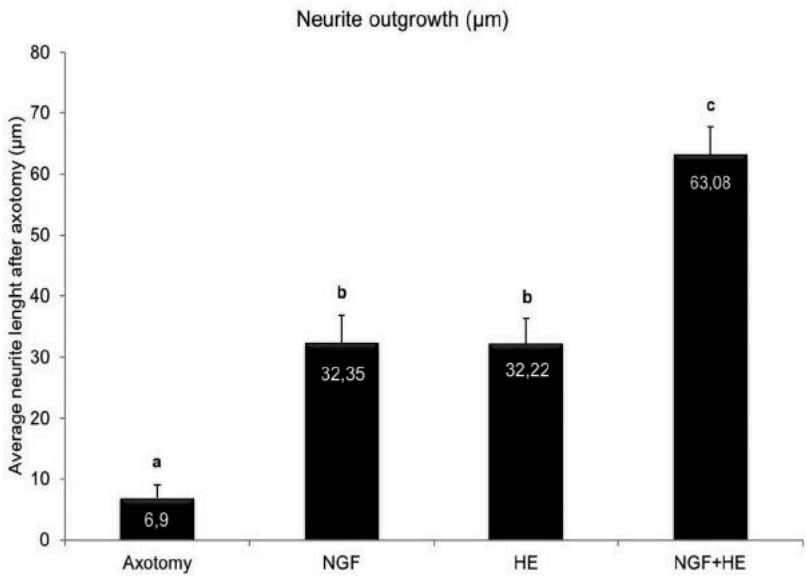

Figure 6. NGF and HE protect neurite (axon) outgrowth against retraction, degeneration and fragmentation of axotomized-axons. Average neurite lengths of the surviving neurons for 48 h after axotomy injury in axotomy group, NGF treatment group, HE treatment group, and NGF+ HE treatment group. Different lower cases represent statistically significant differences ($p < 0.05$).

combination of the NGF and HE treatment also resulted in significantly lengthier axons ($p < 0.05$).

4. Discussion

There is an urgent need to develop effective therapies or to improve existing treatment modalities for PNI [43]. NGF has been reported to promote the maturation and neuritogenesis of developing neurons, the survival and axonal regrowth of injured neurons in the peripheral nervous system [4,8–10]. However, NGF is a rather high molecular-weight protein, it is unable to cross the blood-brain-barrier [11]. Therefore, the potential medicinal values of medicinal mushrooms have become attractive compound in the search for pharmacological remedies for varies pathological conditions including the PNI [13]. HE can promote neuronal survival, initiate nerve regeneration and potentiate neurite outgrowth using the TrkA/Erk1/2 pathway [30,30,44–46]. For this reason, we compared the protective and regenerative effects of NGF and HE or the combination of two compounds on peripheral sensory neurons by using *in vitro* assays, mouse DRG neurons, and laser microdissection techniques. The rodent models of nerve injury have improved our understanding of peripheral nerve regeneration [6]. However, clinical trials have been scarce, in part because such models are not sufficiently recapitulated the *in vivo* process in humans [6]. In human injuries, axons often extend over much longer distances than in mice, and injury leaves distal nerve fibers and target tissues without axonal contact for extended amounts of time [6]. We used a microscope-controlled laser microdissection technic to model axonal injury in mouse DRG neurons under *in vitro* culture conditions [35,37]. This model system provides a preliminary assessment of test compounds and it is also effective, precise, controllable, and easily replicated. The axotomies we performed created minimum traumatic cuts in a pretty clean environment. We have observed that axotomy-induced neuronal cell death was performed via fragmentation in large proportion. In the event of such mechanical injuries, activation of proteolytic enzymes and largely calpains, by calcium entering from inside the axotomy injury is considered to be responsible for the breakdown of cytoskeleton proteins [47]. In addition, it was reported that the axonal fragmentation causes the energy-dependent processes in the cell, the axons become vulnerable leading to their eventual death [35]. Vavilis et al. have reported that Oxygen-Glucose Deprivation lasting for 16 h imposes a severe stress to PC12 cells, causing lethal hypoxia resulting to an only 25% cell survival [48].

We also observed that some neurons reacted to the injury by retracting their neurites, and surviving neurons are composed mostly of these retracting neurons. The retraction process may be due to the preservation of subcellular substrates and energy sources [35]. To survive and preserve energy, an injured neuron may retract its axotomized axon [49] and thus, injured neurons may save a significant amount of energy for protein synthesis from substrates, carry the proteins, regenerate their injured axons, and trigger the energy-dependent processes.

In this study, cells were pretreated with NGF at a final concentration of 50 ng/ml and/or with hot

aqueous extract of HE at 50 μg/ml 48 h prior to axotomy. The literature suggests that the optimal *in vitro* doses for these agents were determined to be 50 ng/ml for NGF [50,51], 50 μg/ml for HE [50]. Therefore, we followed these previously optimized doses in our studies. We have demonstrated that NGF treatment, HE treatment and the combination of both have significant protective and regenerative effects on injured neurons. The protective abilities of HE treatment and its combination with NGF were significantly higher than NGF treatment alone. In addition, regenerative ability of treatment with the combination of both NGF and HE was considerably more potent than single NGF or HE treatment. To the best of our knowledge, for the first time, we observed that the regenerative capacity of combination treatment with NGF and HE was higher than the treatment with individual compounds. Interestingly, the combination of HE and NGF exhibited a superior neurite regeneration ability of axotomized neurons. Our results allow hypothesis that NGF and HE can have different protective and regenerative mechanisms, antioxidative and anti-inflammatory properties. However, high doses of NGF had an inhibitory effect on the neurite growth in DRG explanted from wild-type [52]. Our results obtained on the injured neuron model *in vitro* demonstrated a neuronal survival and neurite outgrowth that are consistent with previous studies [30,32,44,46]. The aqueous extract of HE's fruiting bodies promoted regeneration of the injured peroneal nerve [53] and also increased the rate of recovery in Sprague-Dawley rat models [54]. *In vitro* assays, HE extract was reported to stimulate NGF synthesis and/or promote NGF-induced neurite outgrowth in various cell types [55]. Thus, our results suggest that the HE may be a potent candidate for preventing neuronal death, improving the nerves, treating PNI.

On the other hand, axotomy injury in neurons induces stress and release of free radicals, which lead to the generation of the reactive oxygen species (ROS). The excessive ROS causes damage to the cells in most organisms [28]. Therefore, neutralizing of free radicals or peroxide radicals by an antioxidant agent is essential for cell protection against oxidative stress [28]. HE dose dependently prevented oxidative stress induced by H_2O_2 in GES-1 cells [28,56,57]. The reason that the neuroprotective activity of HE significantly higher than NGF may be associated with the antioxidant and anti-inflammatory properties of HE. Lee et al. showed that oral administration of HE had a protective effect on brain infarction in rats [22]. HE reduced the stroke-induced total infarcted volumes by 22% and 44% in rats [22]. The levels of acute inflammatory cytokines including IL-1β, IL-6, and TNF-α were all reduced by HE [22]. These findings confirm the protective properties of HE mycelium, which include the prevention of ischemic injury to neurons; this protective effect seems to be involved in the *in vivo* activity of iNOS, p38, MAPK, and CHOP [22]. The more potent protective activity of HE may be due to stimulating NGF synthesis and/or promoting NGF-induced neurite outgrowth [55], reducing of oxidative stress [28,29,56], modulating of the anti-inflammatory reaction [22]. We propose that HE can positively affect the survival of neurons by mimicking trophic factors and by reducing oxidative stress and inflammation. Our *in vitro* findings provide the evidence that HE treatment and NGF + HE combine therapy may be utilized in future clinical development. Our studies warrant further investigation to therapeutic functions of the NGF and HE in axotomy-induced PNI. Natural compounds of HE should be further developed to support the possible application of HE in the clinical setting to prevent and treat the common PNI.

5. Conclusions

Although both NGF and HE have the neuroprotective effects on neurons against axotomy-induced cell death, the protective efficacy of HE treatment or the combination of HE with NGF shows a superior protective activity. Regeneration ability of NGF+HE combination is more potent than single compound alone. HE or its combination with NGF may be effective and safe therapeutic option to be utilized in PNI. Therefore, our studies warrant further detailed investigation of NGF and HE as a potential neuroprotective agent to alleviate peripheral nerve degeneration.

Acknowledgments

This study was supported by Van Yüzüncü Yıl University Scientific Research Project Directorate of Turkey [Grant number: 2014-SBEYL148]. The authors thank Dr. Elif kaval Oğuz due to technical support for preparing a primary cell culture from DRG. The authors are also grateful to researcher Ayşe Şeker for organizing the figures.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by the Van Yüzüncü Yıl University Scientific Research Project Directorate of Turkey [2014-SBEYL148].

Notes on contributors

Ramazan Üstün is conducting research to understand the molecular and cellular mechanisms involved in neurodegenerative and neuroregenerative processes using primer

cell culture and animal models as well as advanced microscopic imaging techniques.

Peray Ayhan is a Ph.D. student in the Neuroscience program of Ondokuz Mayıs University. Her research focusses on molecular and cellular mechanisms in mechanical damage model of the peripheral nerve.

ORCID

Ramazan Üstün http://orcid.org/0000-0001-7731-4698

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